Fiscal Year:	FY 2006	**Task Last Updated:**	FY 05/11/2009
PI Name:	Atreya, Arvind Ph.D.		
Project Title:	Extinction of Gaseous Microgravity Diffusion Flames in Various Atmosphere		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous flames - premixed		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	48109-2125	**Congressional District:**	15
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
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No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
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COI Name (Institution):	Im, Hong (University of Michigan)		
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The overall task was divided into experimental and computational parts. The experimental effort consisted of using a porous spherical burner to issue C2H4 fuel and mixtures (with CO2 & N2) into a controlled oxidizing environment (with various concentrations of O2, CO2, and N2). These experiments provided data for understanding flame chemistry and flame radiation interactions. Computational models with detailed chemistry and radiation were developed to better understand these interactions. Only the 2.2s drop tower experiments were completed. The proposed experiments for longer microgravity duration in the 5s drop tower were not completed due to shortened time. The supporting numerical calculations with detailed chemistry have also been completed.
Task Objective:

The objective of this research was to experimentally and computationally investigate the conditions that will lead to

Task Description:	extinction of gaseous diffusion flames in μg and understand the role played by gas and soot radiation in flame extinction. The scientific goal was to develop a fundamental understanding of the interaction between radiation and flame chemistry that is responsible for extinction of diffusion flames. The practical goal was to help develop fire suppression and detection technology for space applications. Task Significance: This research is intended develop a fundamental understanding of the interaction between flame radiation and flame chemistry that is responsible for extinction of diffusion flames. The practical goal is to help develop fire suppression and detection technology for space applications. The results of work are also applicable to practical combustion systems where exhaust gas recirculation is used as a method of pollutant control in engines and furnaces.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research will help develop fire suppression and detection technology for space applications. The results of work and the science developed are applicable to practical combustion systems where exhaust gas recirculation is used as a method of pollutant control in engines and furnaces. Invovative Technologies Developed: 1. Method of transient measurements of spectral radiation emitted from the flame ; 2. Method of determining the spatial soot profile by soot scattering imaging.
Task Progress:	This project was divided into two parts: (i) An experimental component conducted at the NASA-Glenn drop tower facility, and (ii) A computational component conducted at the University of Michigan. Progress made in these components is summarized below: Experimental: The Ph.D. students successfully completed their degree. Dr. Melissa Chernovsky worked on the experimental part of the project and Dr. Songtao Tang worked on the computational part of the project. We have also written several Journal publications and presented conference papers based on their work. These are summarized below. Melissa collected a large amount of data which will be the basis of several future publications. A brief summary of the work is presented below: Experimental work: Melissa's Ph.D. thesis presents experimental research conducted to improve our understanding of flames in microgravity conditions. Of particular interest is to understand the effects of radiation heat loss and reabsorption because of their importance in flame extinction and implications for fire safety in low gravitational environments. Spherical diffusion flames were chosen for this study because they amplify the effect of radiation heat loss. Flames in microgravity naturally develop a spherical character due to diffusional symmetry, and consequently their one-dimensionality enables detailed models with chemistry and radiation. The focus of this microgravity diffusion flame study is to examine the effects of various diluents introduced on the fuel and oxidizer sides of a spherical diffusion flame. Characterization of the flame consisted of flame growth rate, temperature field, and spectral radiation emission measurements. Transient measurements were conducted at atmospheric pressure using a porous spherical burner aerodynamically supporting an ethylene diffusion flame in an oxidizing atmosphere. The effect of diluent on the fuel side of the reaction zone was investigated by comparing non-radiative N2 with a radiative diluent mixture including CO2 and He, while maintaining equivalent adiabatic flame temperatures. The influence of CO2 on the fuel side of the reaction zone reduced flame temperatures and inhibited soot formation. CO2 enhanced the heat release rate of the flames without contributing significantly to radiation reabsorption. The effect of diluent on the oxidizer side of the reaction zone was similarly investigated by replacing N2 in the oxidizer with CO2 and/or He. The influence of He on the oxidizer side initially increased heat release of the flame and decreased the flame temperature due to the enhanced thermal conductivity dominating the decreased thermal capacity of the mixture. The influence of CO2 on the oxidizer side also reduced flame temperatures and inhibited soot formation. However, the mechanisms by which CO2 affected the oxidizer were different from that of the fuel side. While CO2 in the oxidizer similarly enhanced heat release from the flame, spectral radiation emission measurements showed significant radiation reabsorption effects. This study provides important implications for extinguishment mechanisms of flames in low gravitational environments. Computational Work: The goal of the computational work was to assess the effects of gas radiation as a means to promote flame extinction. To investigate these phenomena, a one-dimensional computational model was developed to simulate the evolution of a spherical diffusion flame with consideration of detailed chemistry and transport properties. Various levels of radiation models were implemented and the results were compared with experimental measurements of flame radius and temperature profiles. It was shown that the statistical narrow band (SNB) combined with the discrete ordinate method (DOM) reproduced the experimental results with highest accuracy, and this combination of the radiation models were adopted in the subsequent parametric studies. The parametric studies explored the relative effectiveness of fuel- and oxidizer-side dilution on the flame radius and temperature behavior, with nitrogen, CO2, and helium as diluents. In the spherical configuration considered in this study, the oxidizer-side dilution has a stronger effect on flame transient behavior than the fuel-side dilution, thereby suggesting a more effective means to induce flame extinction by dilution. Study on various oxidizer-side dilution cases shows that CO2 has a larger suppression effect than helium or nitrogen with the same dilution level. CO2 dilution has multiple effects on flame behavior including radiation, thermodynamic, diffusion, and chemical effects. Quantitative analysis shows that the radiation effect is the primary factor accounting for flame temperature drop by approximately 60%, as compared to the thermal/diffusion (30%), and chemical effect (10%). Considering the dominance of the radiative heat loss on flame extinction, a unified extinction criterion that applies to a wide range of parametric conditions was sought. The compiled computational results indicated that a critical flame temperature of 1130 K at extinction appears to be valid for most of the conditions under study. Therefore, it is concluded that extinction of spherical diffusion flame is primarily dictated by the local condition in the flame zone rather than by the volumetric radiative heat transfer in the surrounding gases.

Bibliography Type:	Description: (Last Updated: 04/21/2017)
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